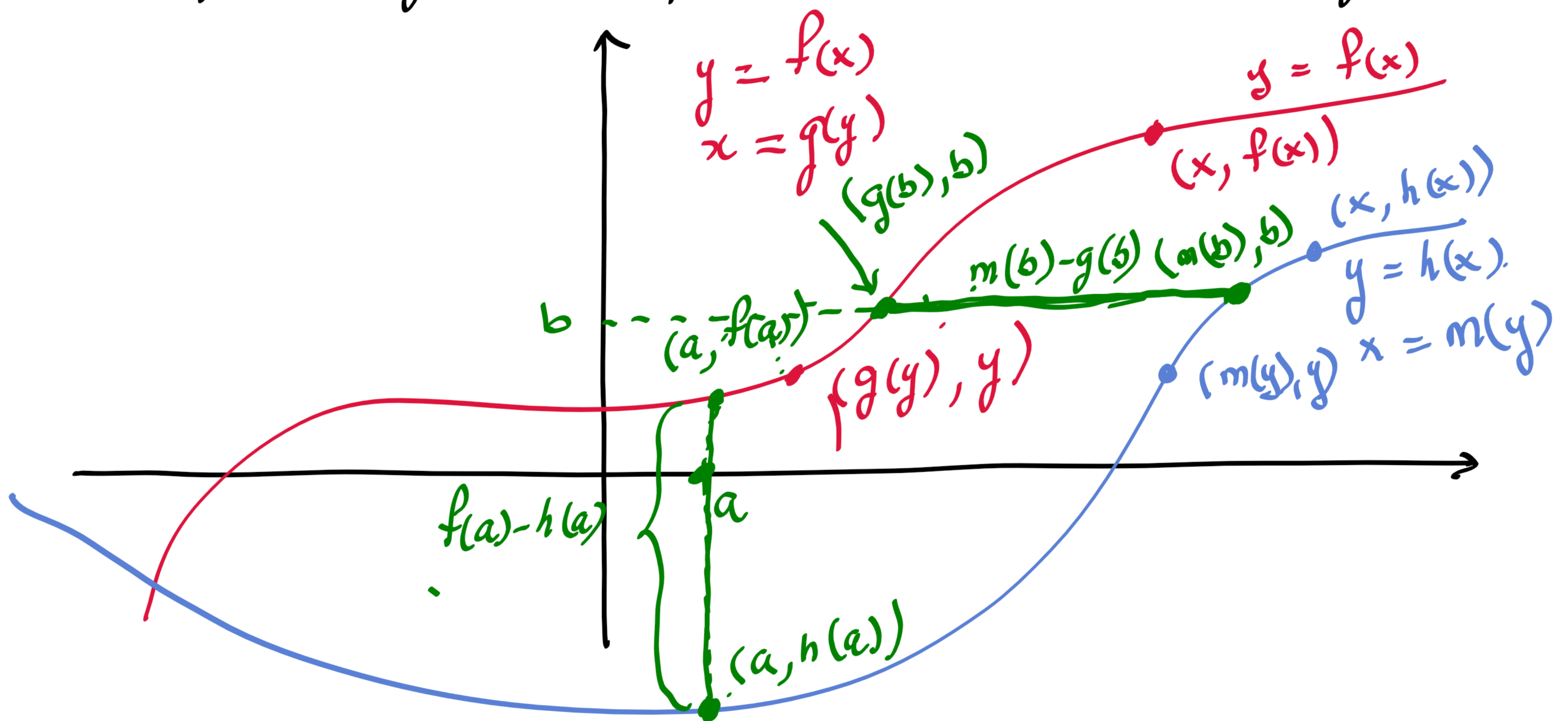
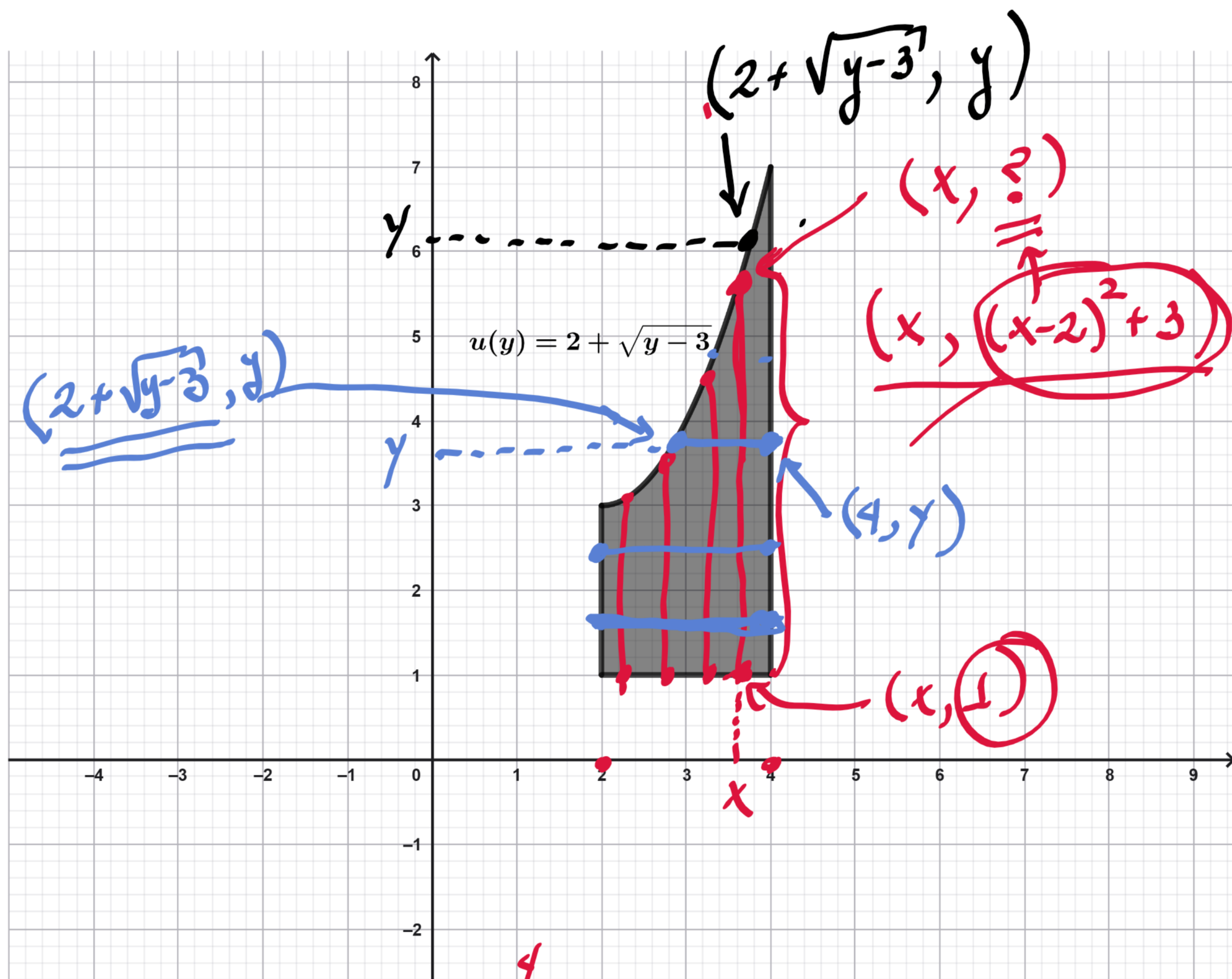


Distance from the graph of a function to the x -axis and y -axis.





$$x = 2 + \sqrt{y-3}$$

$$x-2 = \sqrt{y-3}$$

$$(x-2)^2 = y-3$$

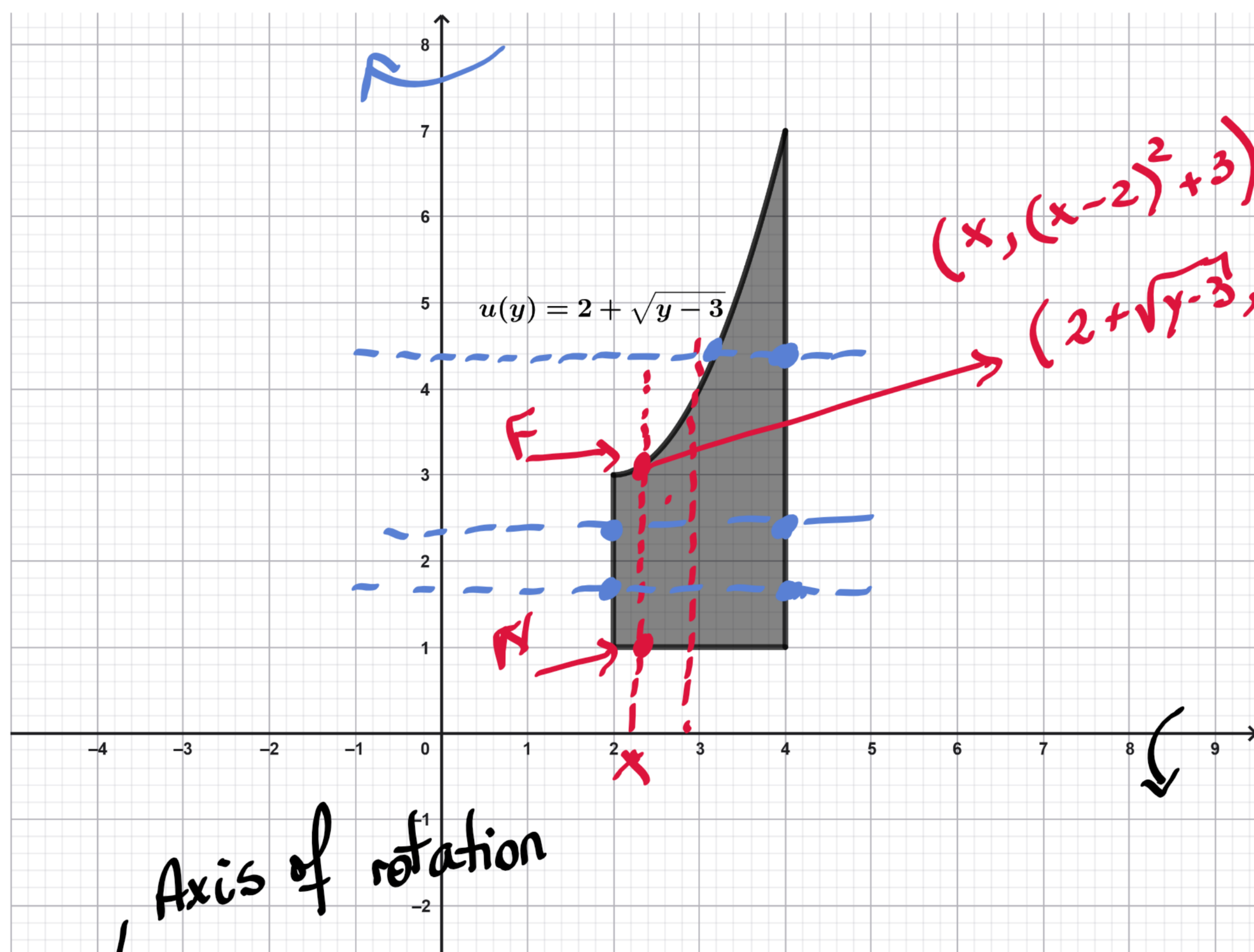
$$(x-2)^2 + 3 = y$$

Area of the region

$$A = \int_2^4 ((x-2)^2 + 3 - 1) dx$$

$$A = \int_1^3 2 dy + \int_3^7 (4 - (2 + \sqrt{y-3})) dy$$

← Axis of rotation



x-axis:

$$r(x) = 1$$

$$R(x) = (x-2)^2 + 3$$

$$\int_2^4 \pi (R(x)^2 - r(x)^2) dx$$

$$= \int_2^4 \pi ((x-2)^2 + 3)^2 - 1 dx$$

Axis of rotation
y-axis:

$$[1, 3]$$

$$R(y) = 4$$

$$r(y) = 2$$

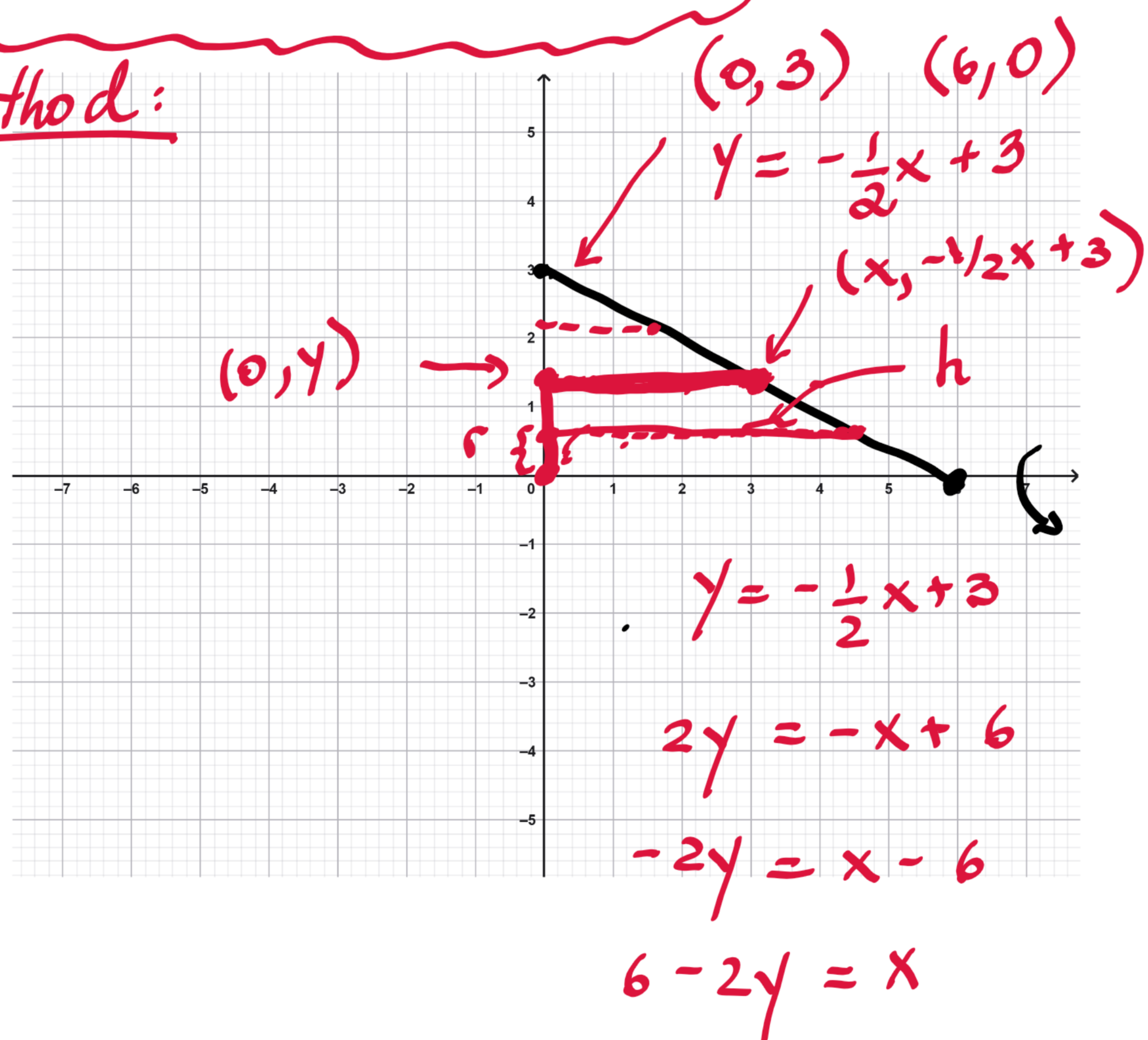
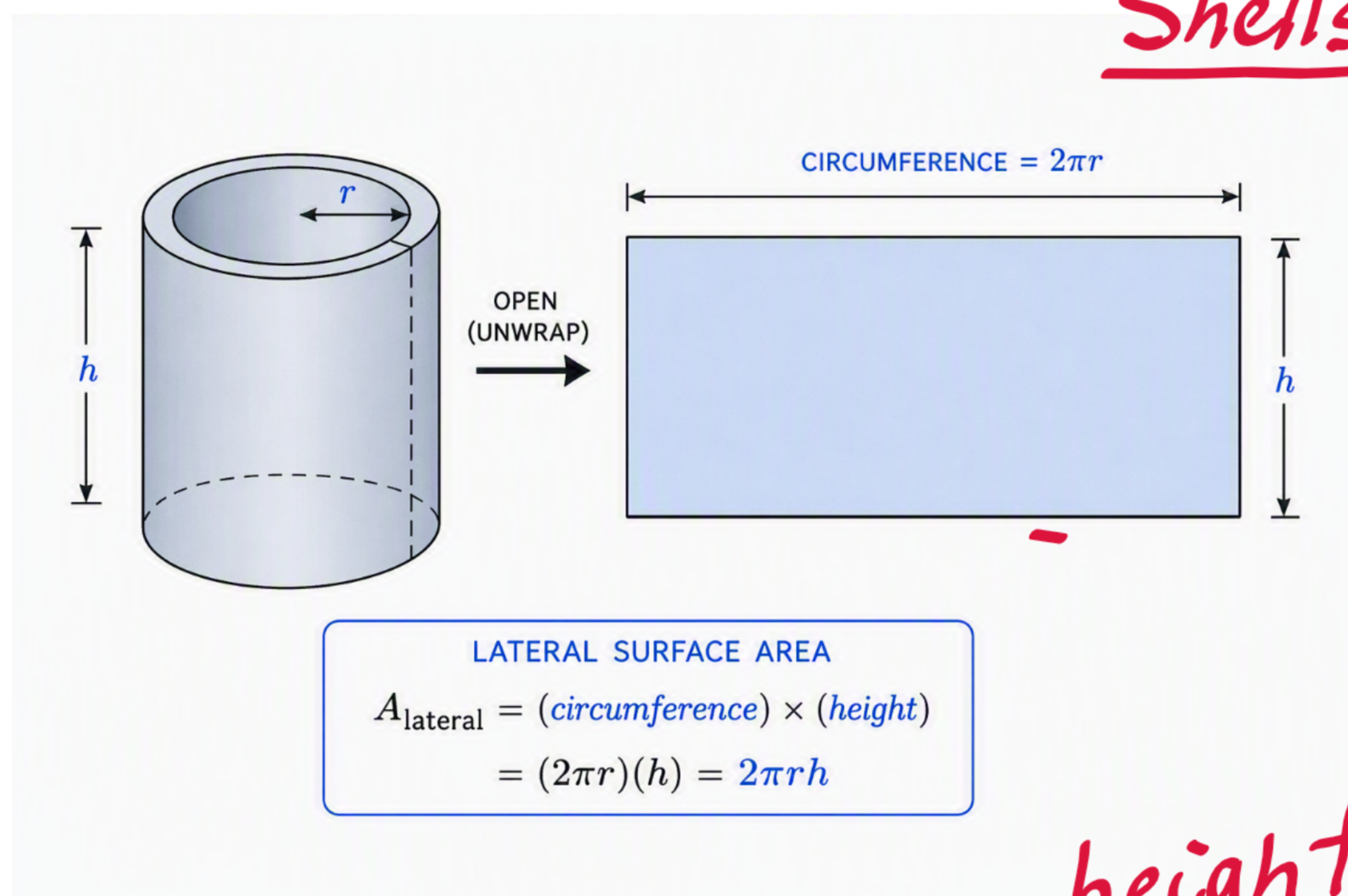
$$[3, 7]$$

$$R(y) = 4$$

$$r(y) = 2 + \sqrt{y-3}$$

$$V = \int_1^3 \pi (16 - 4) dy + \int_3^7 \pi (16 - (2 + \sqrt{y-3})^2) dy$$

Shells method:



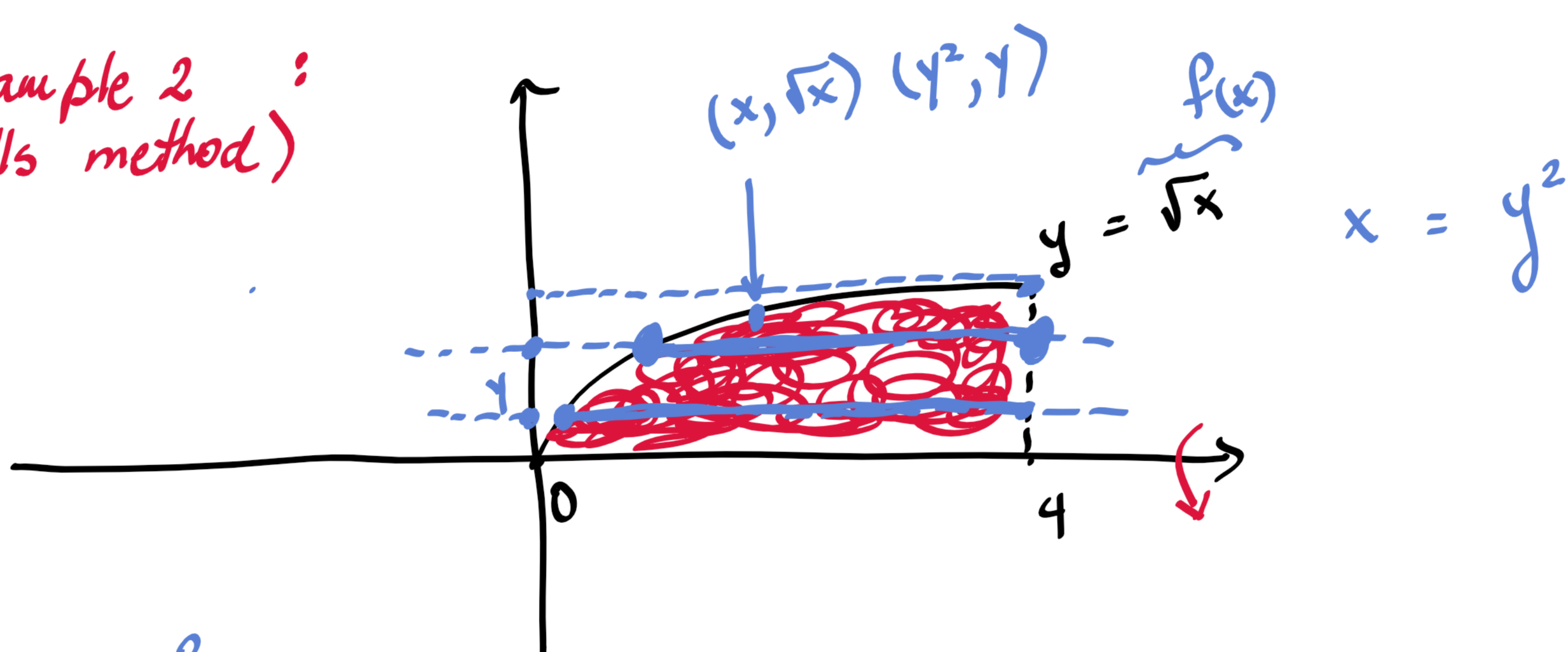
$$\int_0^3 \underbrace{2\pi(r)}_{\text{radius}} \underbrace{(6-2y)}_{\text{height}} dy$$

$$2\pi \int_0^3 \underbrace{(6y - 2y^2)}_{2(3y - y^2)} dy = 4\pi \int_0^3 (3y - y^2) dy$$

$$= 4\pi \left[\frac{3}{2} y^2 - \frac{1}{3} y^3 \right]_0^3$$

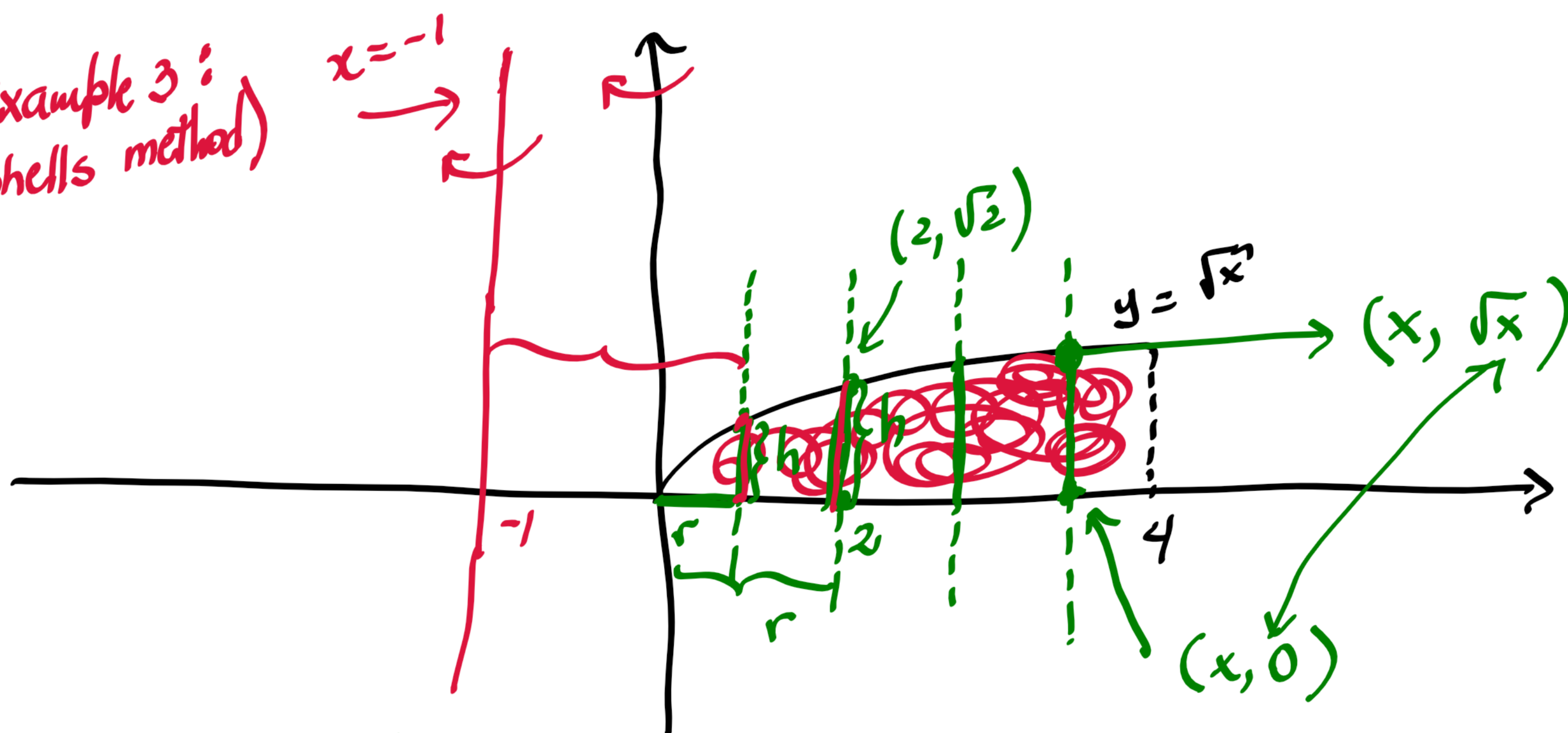
$$= 4\pi \left[\frac{27}{2} - \frac{27}{3} \right] = 18\pi.$$

Example 2:
(Shells method)



$$V = \int_0^2 2\pi \underbrace{(\quad y \quad)}_{\text{radius}} \underbrace{(\quad 4 - y^2 \quad)}_{\text{height}} dy$$

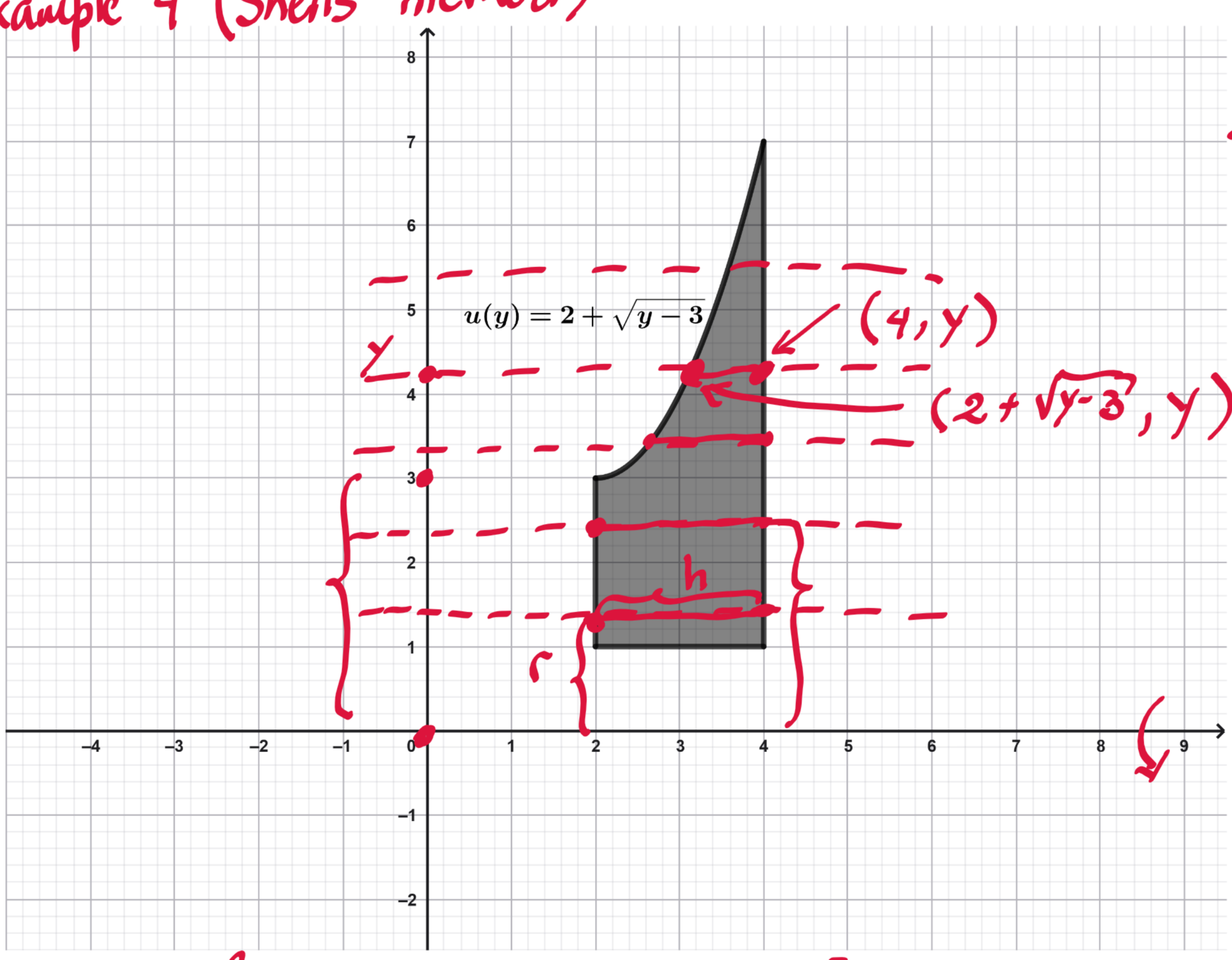
Example 3:
(Shells method)



$$V = \int_0^4 2\pi (\quad x \quad) (\quad \sqrt{x} \quad) dx$$

$$V = \int_0^4 2\pi (x+1)(\sqrt{x}) dx \quad (\text{Axis of rotation is } x=-1)$$

Example 4 (Shells method)

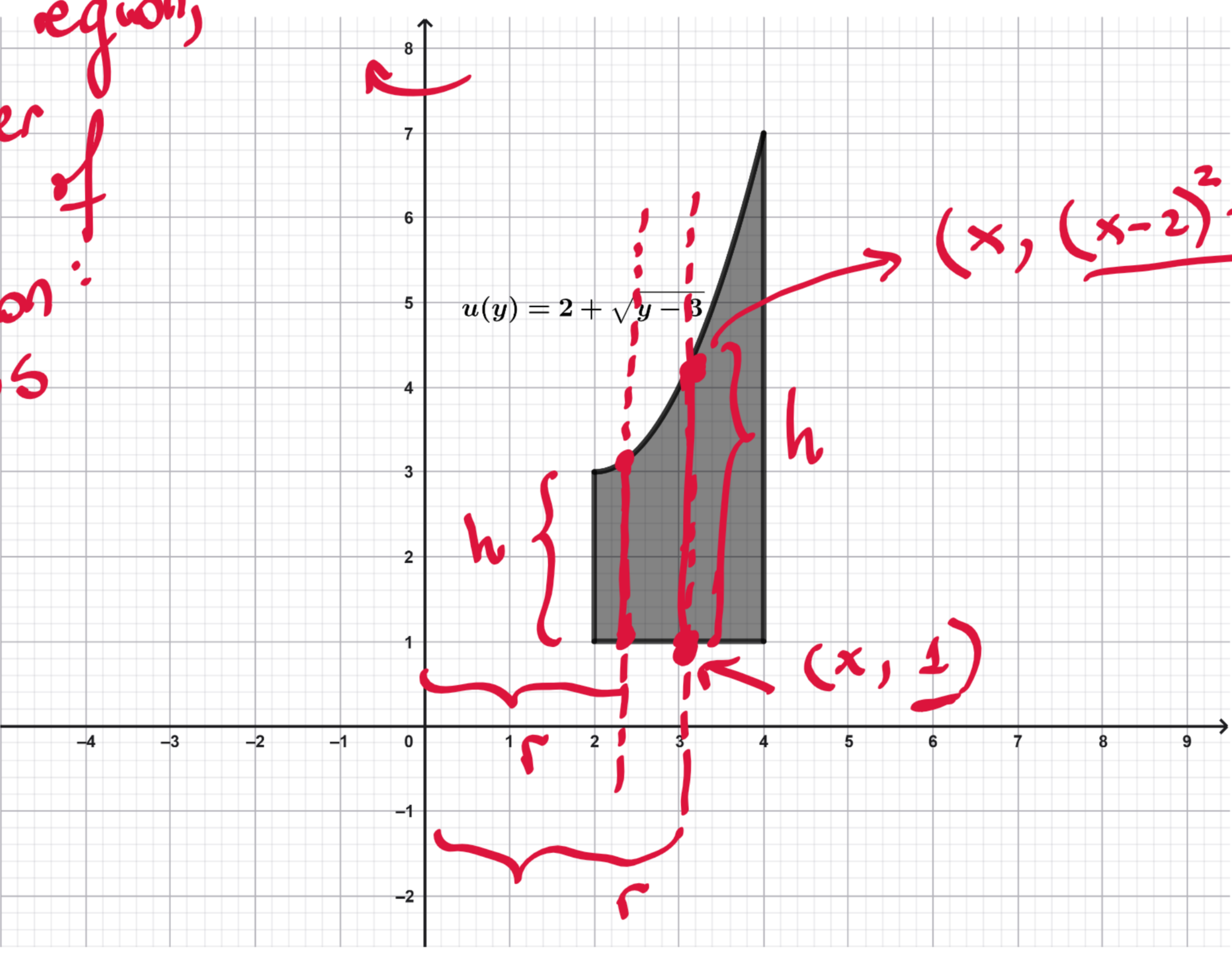


$$\begin{cases} x = 2 + \sqrt{y-3} \\ y = (x-2)^2 + 3 \end{cases}$$

Axis of rotation: x-axis

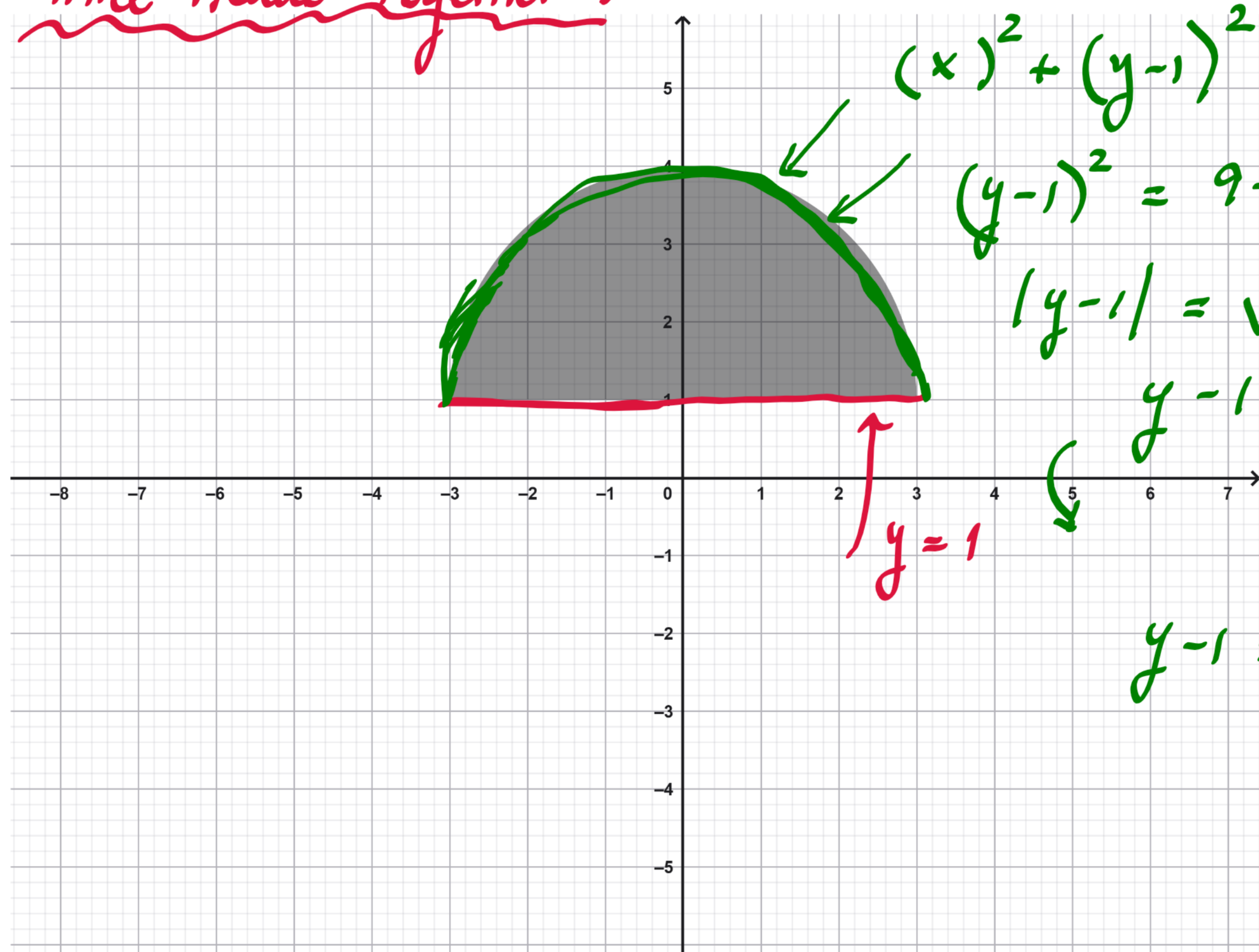
$$V = \int_1^3 2\pi(y) \cdot 2 \, dy + \int_3^7 2\pi(y) (4 - (2 + \sqrt{y-3})) \, dy$$

Same region,
another
axis of
rotation:
y-axis



$$V = \int_2^4 2\pi(\underbrace{x}_{\text{radius}}) (\underbrace{(x-2)^2 + 3 - 1}_{\text{height}}) \, dx$$

Three Heads Together:



$$(x)^2 + (y-1)^2 = 9$$

$$(y-1)^2 = 9 - x^2$$

$$|y-1| = \sqrt{9-x^2}$$

$$y-1 = \sqrt{9-x^2} \Rightarrow y = 1 + \sqrt{9-x^2}$$

or

$$y-1 = -\sqrt{9-x^2} \Rightarrow y = 1 - \sqrt{9-x^2}$$